

# Discovering Exoplanets and Their Properties

Using Data from Adler Planetarium and NASA

Astronomy 1126 Project 2

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December 10, 2025

## **Abstract**

In this study, we will compare observations of exoplanets from data collected by NASA and deep space telescopes, including the Kepler Space Telescope, launched in 2009. Our largest body of data will be from light curves available publicly online within a variety of astronomical databases. These include the Mikulski Archive for Space Telescopes (MAST), data from Andrew Vanderburg, an astronomical researcher at Harvard, and from EXOTIC, a software used by multiple organizations for storing and analyzing data on exoplanets. To create scatter plots of these curves, we will use Microsoft Excel to import the data and use several functions in an attempt to find the minimum and maximum magnitudes of stars, along with the mass, orbital radius, planetary radius, and density of the exoplanets present. Our goals are to determine different properties of exoplanets to estimate whether any of them are within the habitable zone of their stars and what other properties may or may not give them the potential to support forms of life.

## **Introduction**

As children, we gaze into the abyss, wondering what lies out there. Are we alone in this vast universe? Is our solar system the only one that has planets? Is ours the only one that carries life? These questions, seemingly simple yet profound, are asked by children, philosophers, and scientists throughout history. While once thought impossible, in the last three decades, astronomers have discovered thousands of exoplanets, with a total of more than 6,000 at the time of writing.

Some small, some large. Some close to their stars or far away. Some cold or hot, some dense, some not. In the decades since exoplanets have been confirmed, astronomers have learned a few things. The first is that the vast majority of stars have planets that orbit them. This means we have only scratched the surface of a tiny, tiny percentage of planets for us to find. The next is that there are many planets that reside within the habitable distance from their star, and there are many that appear to have the potential to contain water, either in a frozen or a gaseous state.

There are many databases out there with information open to the public. Many of them have catalogs of data from thousands of stars observed over time by space satellites. There is a lot of

encouragement from the hosts of these archives for members of the public to analyze this data and publish results. One of our goals is to see how easy or difficult it is for the public to find and utilize this data. With little to no prior experience, is it feasible for untrained individuals to create light curves and perform calculations from the data readily available? With a limited amount of time, are we able to produce accurate calculations? The goal of this analysis, as amateur astronomers, is to learn about established methods of detection and to see if we can utilize them ourselves.

We will begin by observing data from exoplanets that have been confirmed by the scientific community. This will give us a baseline of what to look for. The next step will be to look at data sets of potential exoplanets that have not yet been confirmed. It is our goal to attempt to predict how likely or unlikely a dataset is to contain evidence of an exoplanet. Finally, we will analyze the data we collected to try to determine as many properties of the exoplanets or planetary systems as possible.

### **Observations and Data Acquisition**

Our first portion of data comes from exo.MAST. MAST is a collection of astronomical observations made by space telescopes. The general database has information on all stars regardless of whether they contain an exoplanet or not. The database, exo.MAST contains data only if a star system either has a confirmed exoplanet or is a candidate waiting to be confirmed or rejected. This helps narrow things down tremendously. To search for a specific planet within “exo.MAST,” it is most helpful to use a planet's “TIC” number. One example of a confirmed exoplanet found this way is TIC 55525572 b. Notice the letter “b” at the end. If it has a letter, it is a confirmed planet; if it does not have a letter, it is unconfirmed.

Exo.MAST has several tools with helpful information about exoplanets listed. These include transit light curves compiled from data by TESS (the Transiting Exoplanet Survey Satellite). Once a planet is selected, there is a tab somewhere on the webpage that should allow us to see the light curve taken by the telescope. These are normalized and folded on top of each other, meaning multiple transits have been stacked one on top of another as opposed to having them in a very, very long line. There can sometimes be more than 20 transits on a single graph numbered

consecutively. The odd and even numbered transits are divided into two separate graphs. If both graphs look the same, it means the transit is indicative of an exoplanet. If the left and right graphs are different, it may indicate the occurrence of more than one star, a twin star system.

Our next set of data comes from Andrew Vanderburg, an astronomical researcher at Harvard. He took data from the Kepler Space Telescope and made adjustments to make it more useful. This data is easily downloadable as a text file and able to be opened in Microsoft Excel. Once in Excel, we took the entire data set and made a scatter plot, or in other words, a light curve. With this dataset, we used formulas to find the minimum and maximum brightness of the star as its planet completed its transit. From there, we found the transit depth (the difference between max and min) and combined that in a formula with the known radius of the planet's star to calculate the radius of the planet itself.

| Earth For Comparison |                          |                   |                   |
|----------------------|--------------------------|-------------------|-------------------|
| Radius               | Distance From Star (Sun) | Star Radius (Sun) | Orbit Length      |
| 3,959 miles          | 93 million miles         | 865,000 miles     | 1 year (365 Days) |

Figure 1 - Planet Earth Data

| Jupiter For Comparison |                          |                   |                        |
|------------------------|--------------------------|-------------------|------------------------|
| Radius                 | Distance From Star (Sun) | Star Radius (Sun) | Orbit Length           |
| 44,423 miles           | 484 million miles        | 865,000 miles     | 12 years (Jovian Year) |

Figure 2 - Planet Jupiter Data

| Mars For Comparison |                          |                   |              |
|---------------------|--------------------------|-------------------|--------------|
| Radius              | Distance From Star (Sun) | Star Radius (Sun) | Orbit Length |
| 2,106 miles         | 142 million miles        | 865,000 miles     | 687 days     |

Figure 3 - Planet Mars Data

| EPIC 211311380 |                |             |                    |               |
|----------------|----------------|-------------|--------------------|---------------|
| Star           | Exoplanet      | Max Flux    | Min Flux           | Transit Depth |
| HIP 41378      | HIP 41378 f    | 1.0021      | 0.9963             | 0.0058        |
| Planet Radius  | Planet Density | Star Radius | Distance From Star | Orbit Length  |
| 49,604 mi      |                | 631,345 mi  | 127,410,000 mi     | 542 Days      |

Figure 4 - Data for Planet HIP 41378 f

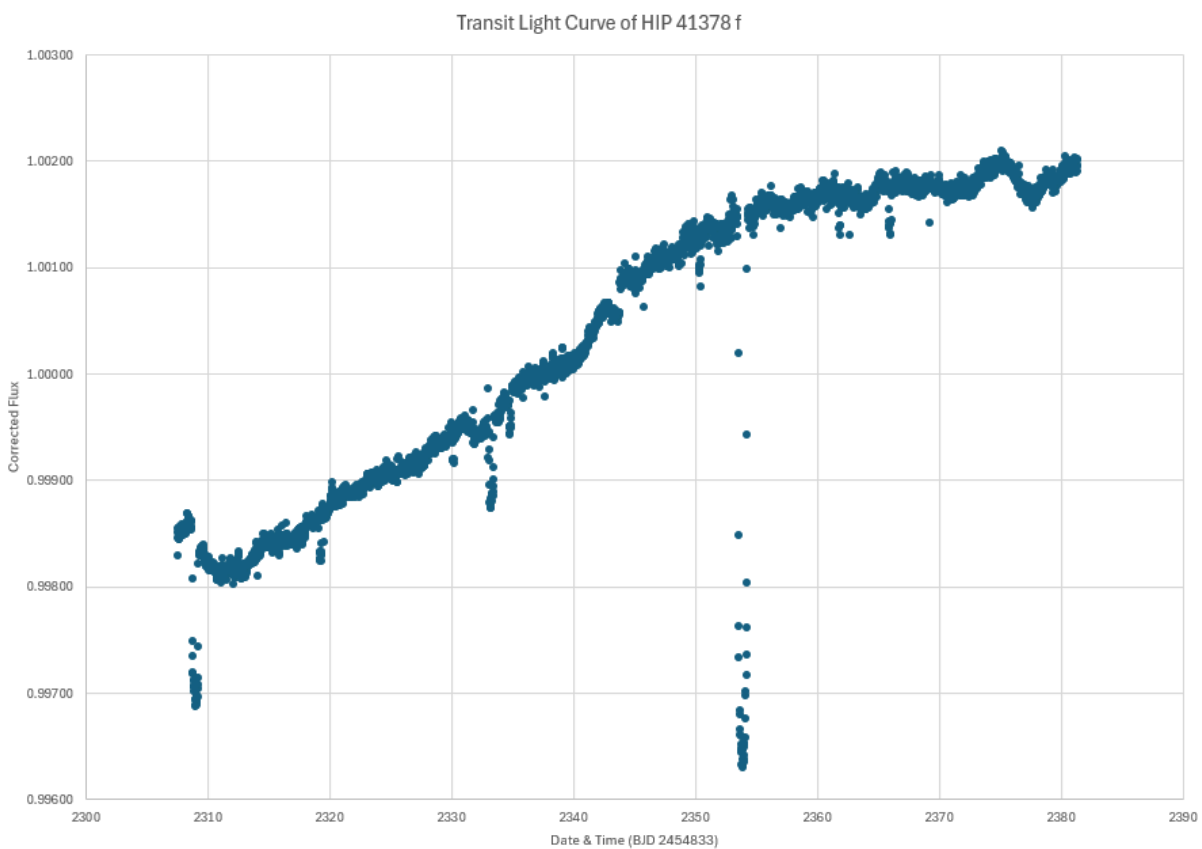


Figure 5 - Light curve generated in Microsoft Excel based on data from Vanderburg for star HIP 41378

| EPIC 201912552 |                |             |                    |               |
|----------------|----------------|-------------|--------------------|---------------|
| Star           | Exoplanet      | Max Flux    | Min Flux           | Transit Depth |
| K2-18          | K2-18 b        | 1.00955     | 0.9914             | 0.01814       |
| Planet Radius  | Planet Density | Star Radius | Distance From Star | Orbit Length  |
| 27,310 mi      |                | 202,743 mi  | 13,292,680 mi      | 33 Days       |

Figure 6 - Data for Planet K2-18 b

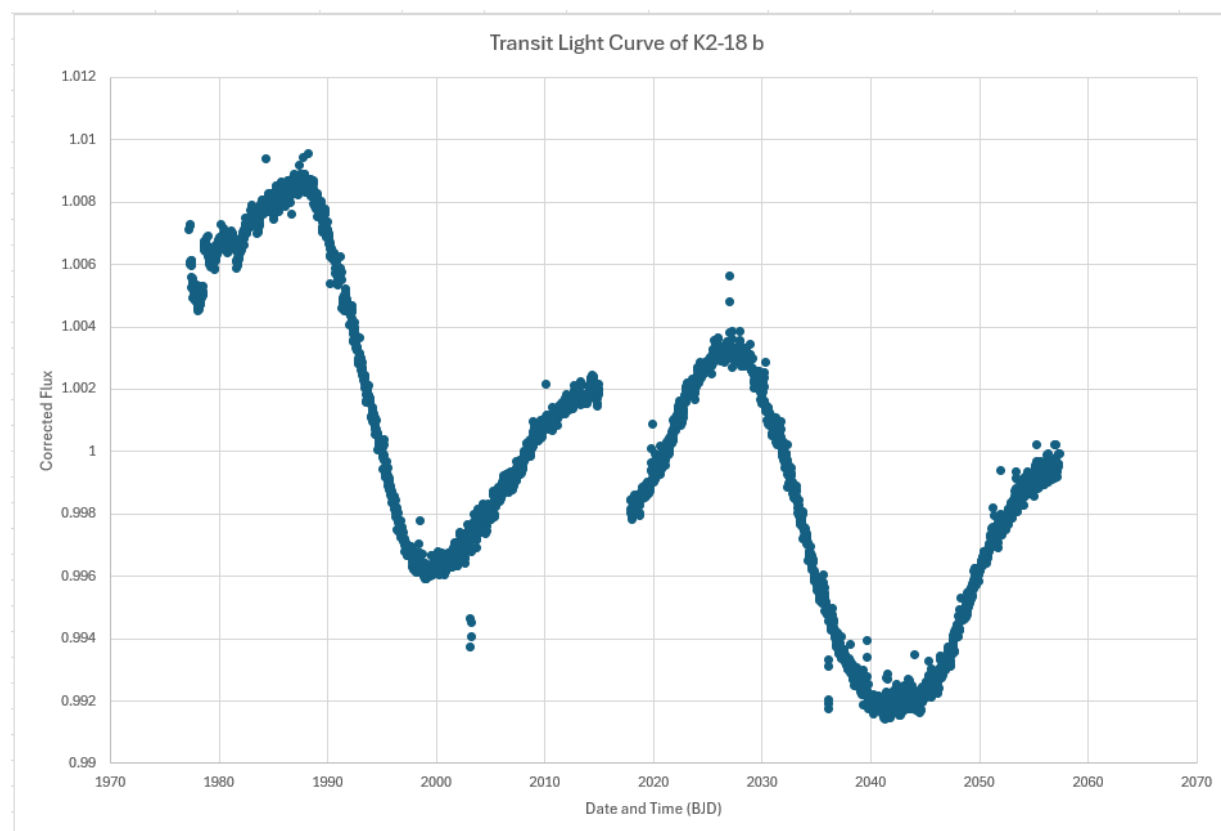


Figure 7 - Light curve generated in Microsoft Excel based on data from Vanderburg for star K2-18

Another research objective of ours was to gain hands-on experience with astronomy by using raw telescope images from Adler Planetarium's Doane Observatory, targeting the star WTS-2 and potential exoplanet WTS-2 b. Using NASA's ExoWatch program, we identified the data and processed it using EXOTIC software in Google Colab, a Python-based coding notebook. We first attended Professor DalSanto's and Professor Nault's lectures to learn the basics of exoplanet observation. Then, we attended a Zoom session with Adler Planetarium and the College of DuPage to understand how to use their data. Adler provided a Google Drive folder containing 165 .fits extension images and calibration frames (biases, darks, and flats). We mounted this folder into NASA's preset Google Colab standard notebook for Exoplanet research. We processed the data using EXOTIC, which measured the brightness of the star and compiled the measurements over time to generate a light curve. In EXOTIC, we had to find and enter the coordinates of the target star, two comparison stars, the latitude, longitude, and altitude of Adler's Doane Observatory telescope. After a painstaking process of troubleshooting error codes, writing Python scripts, and switching to NASA's exoplanet Google Colab notebook, we finally were able to generate the light curve. The images were collected using Adler's PlaneWave CDK24, which is the largest public telescope in the Chicagoland area. It gathers over 7,000 times more light than the human eye and is equipped with research-grade instruments for photometry of stars, exoplanets, asteroids, and other celestial objects, even in Chicago's downtown light pollution. This process helped give us hands-on experience using actual NASA-created programs and actual powerful telescopes operated by high-level observatories, illustrating how astronomers transform raw images into analyzable data.



Figure 9 - EXOTIC Google Colab Star Map

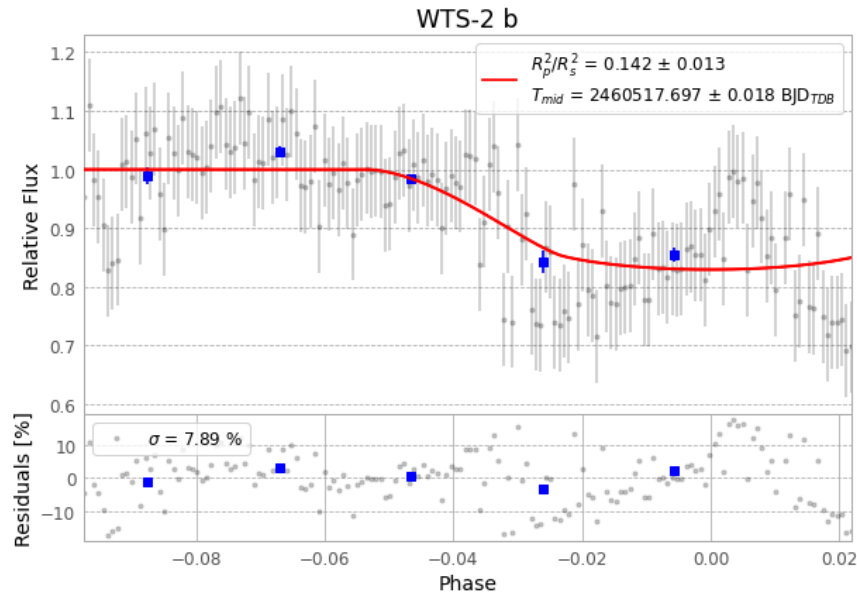


Figure 10 - EXOTIC Adler Dataset WTS-2 lightcurve

### Data Analysis and Methodology

The first step in analyzing our data was opening the files in Microsoft Excel. The files we used are of data from the Kepler Space Telescope on its K2 mission. These are publicly available, and each one has thousands of data points measuring the changes in observable brightness of stars over time. Separating the data into two columns, column one for date and time, column two for brightness, we were able to use Excel to convert the data into a light curve. Once the light curves were created, we were able to use formulas to determine several properties about the star and the exoplanet within the different planetary systems we observed.

The first step is determining brightness, also known as flux. To do this, we selected all of the data for levels of brightness and had Excel give us the highest value, the maximum brightness, or maximum flux. In the same way, we used a formula to have Excel give us the lowest value, or minimum flux. From there, we subtracted the minimum from the maximum to give us the difference between the two values, also known as the transit depth.

For our next set of calculations, we needed a piece of information we were not able to acquire from the light curves alone. For this, we needed to rely on data publicly available, specifically

the radius (size) of the stars these planets were orbiting. Once acquired, we were able to plug the size of these stars combined with the transit depths into the following formula.

$$\text{Planet Radius} = \text{Star Radius} \times \sqrt{(\text{square root of}) \text{ Transit Depth}}$$

Our next source of data is looking at pre-generated light curves from exo.MAST, a database of confirmed exoplanets as well as likely candidates. One of the things we are able to analyze from this kind of data appears in the light curves. The light curves from exo.MAST has the transits for each planet stacked on top of each other. In addition, the transits are divided into two groups. Odd-numbered transits are stacked on top of each other on the left, and even-numbered transits. If the left and right curves match, it is a strong indication that a planet is what we are observing. If the left and right sides are uneven, one has a depth greater than the other, then it is a possible indication that we are observing a system with multiple stars rather than just an exoplanet.

Additionally, there is one other source of data that can be acquired from exo.MAST was useful for our analysis; a summary page with a variety of data collected on the exoplanet and the planetary system it resides in. This includes light curves, but also the radius, orbital period, and other data related to the exoplanet as well. Anything in red is flagged by the system as unexpected, drawing the legitimacy of this being an exoplanet into question. The data points in red need to be checked before the exoplanet is considered confirmed. Finally, the last piece of helpful information from exo.MAST is the *Difference Image Offset* chart. The suspected planet is in red, at the center of the pink crosshairs. If the red circle is in the middle of the large blue circle, it is a strong indication that what we are looking at is indeed an exoplanet. If the red circle is off to the side outside of the blue circle, it may indicate we are observing something else.

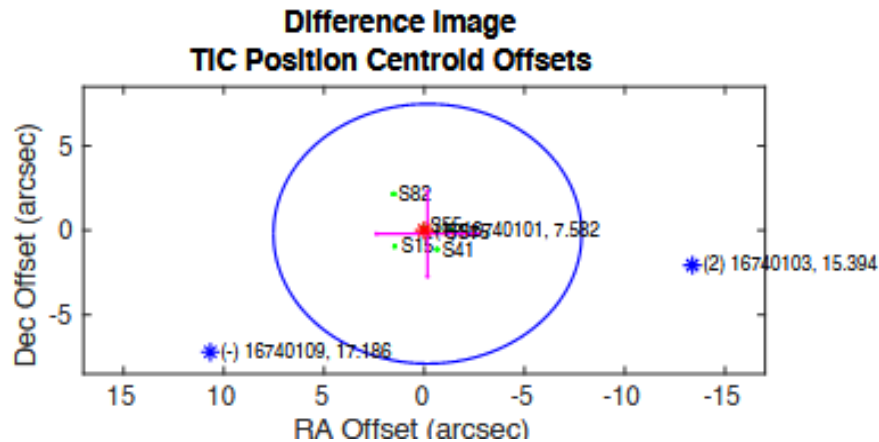


Figure 11 - Difference Image

| DV Fit Results:                 | DV Diagnostic Results:                  |
|---------------------------------|-----------------------------------------|
| Period = 1.48112 [0.00000] d    | ShortPeriod-sig: 0.0% [0.00 sigma]      |
| Epoch = 1683.4453 [0.0000] BTJD | LongPeriod-sig: 99.8% [3.05 sigma]      |
| Rp/R* = 0.0809 [0.0001]         | ModelChiSquare2-sig: 100.0%             |
| a/R* = 3.10 [0.01]              | ModelChiSquareGof-sig: 100.0%           |
| b = 0.34 [0.01]                 | Bootstrap-pfa: 0.00e+00                 |
| Seff = 33325.87 [3973.77]       | GhostDiagnostic-chr: N/A                |
| Teq = 3446 [103] K              | OotOffset-rm: 0.112 arcsec [0.04 sigma] |
| Rp = 20.19 [0.52] Re            | TicOffset-rm: 0.264 arcsec [0.10 sigma] |
| a = 0.0342 [0.0018] AU          | OotOffset-tot: 6                        |
| Rho = 0.182 [0.002]             | TicOffset-tot: 6                        |
| Ag = 0.71 [0.07] [-3.96 sigma]  | DiffImageQuality-fgm: 1.00 [6/6]        |
| TP = 4894 [90] K [10.59 sigma]  | DiffImageOverlap-fno: 0.00 [0/6]        |

Figure 12 - Diagnostic Results, Red Indicates Unconfirmed Data

## Results

The easiest property of planets to determine is whether or not they are in the Goldilocks zone. The most common solution for discovering exoplanets is using the transit method. This measures the reduction in light that occurs when planets pass in front of the stars they orbit. With modern space telescopes, it is possible to measure minute changes in reductions of brightness as exoplanets pass in front of their stars, often 1-3% differences. This allows us to easily determine the orbital radius, which in turn tells us how far or close a planet is from its star. Therefore, this information tells us exactly whether or not an exoplanet is or is not within its habitable zone.

Exoplanet K2-18 b is only 13 million miles away from its star. This is 80 million miles closer than the Earth, which is 93 million miles away from our Sun. This likely makes it much, much

hotter than it is here and likely unsuitable to support life as we know it. K2-18 b is not in the habitable zone. The next exoplanet we looked at is HIP 41378, which is 127 million miles away from its star. This is much closer to Earth's distance and puts it within the predicted habitable zone.

While at first glance it may seem all that needs to be done is to compare the orbital radius of these planets to the orbit of Earth, there is one more factor that needs to be taken into account. That factor is the temperature of the stars these planets are orbiting. The star HIP 41378 is roughly 6,371 K. This is quite close to the temperature of the sun at 5,772 K. This combination of star temperature and distance makes HIP 41378 a good candidate for a chance at supporting life. The temperature of star K2-18 is much colder at only 3,650 K. Even though the star itself is much colder, due to the extremely proximity of the planet's orbit, it makes it highly unlikely for K2-18 to be capable of supporting any kind of life that we know of.

The other properties of planets that are easy to determine are planetary radius by using a formula that combines the radius of the host star with the flux depth (difference between maximum and minimum star magnitudes as the planet passes in front of it). The formula is as follows:

$$\text{Planet Radius} = \text{Star Radius} \times \sqrt{\text{Transit Depth}}$$

Based on our observations, we estimate the radius of K2-18 b to be approximately 27,310 miles across. Our estimated radius of HIP 41738 is 49,604 miles. Both of these are substantially larger than the radius of our Earth, which measures close to 4,000 miles. This makes sense, as the larger a planet is, the easier it is to detect. This is due simply to larger planets blocking more light from their star when they complete their transit.

## **Discussion**

One of the hardest portions of this analysis was finding data to work with. There are numerous sites on the internet with thousands of collections of data points, images, and other information on these planets and stars. Most of the sites use differing naming systems when trying to look up specific planetary systems. They also differ in the software needed to analyze and view their

data. Some data sources are easily manipulated in Microsoft Excel. Others require a level of knowledge with programming languages, most often Python, a powerful language known for its simplicity, capability, and far-reaching ability. This makes it difficult for those not acquainted with Python to view the data properly. Despite the abundance of data out there, figuring out how to utilize it was extremely challenging. Much more difficult than aiming and looking through a physical telescope.

Once the data was collected, however, most of the calculations were fairly straightforward. The difficulty here is that many calculations are built on top of each other. They rely on answers from the first one, and if the first answer is incorrect or skewed, it may throw the rest of the calculations off by orders of magnitude. The first calculations in most cases consist of the minimum and maximum levels of flux (magnitudes) of stars, and subsequently the transit depth (the difference between the minimum and maximum flux levels). From there, you can continue with the calculations of planetary radius, orbital radius, and other measurements.

## **Conclusion**

This project demonstrated how publicly available data, along with hands-on observational experience, can, in fact, be used by amateur astronomers and students to estimate key exoplanet properties such as radius, orbital distance, and potential habitability. Our analysis of NASA and exo.MAST data allowed us to determine that K2-18 b is too close to its star to support surface water or life, while HIP 41378 f lies within the habitable Goldilocks zone. The Alder Planetarium data provided practical experience in generating light curves from raw telescope images for data calibration, photometry, and analysis purposes in professional astronomy. Overall, this project allowed us to explore how even amateur astronomers, with proper tools, knowledge, and guidance, can meaningfully explore exoplanets and gain insights. So, to answer the questions asked by children, philosophers, and astronomers alike, is there any evidence of life elsewhere in space? The answer is, we don't know, but according to NASA, there are over 6,000 exoplanets in which there could one day be life.

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